# NUR HIJRAH AS SALAM AL IHSAN

## Applied AI Engineer | Senior Full-Stack Software Engineer

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### PROFESSIONAL PROFILE

Applied AI Engineer and full-stack software engineer with 5+ years of experience designing and delivering production systems. Founder of AsistenDiet and BundaGizi, building production AI-powered products that connect LLM workflows to real-world services through WhatsApp and web applications. Experienced in backend architecture, APIs, databases, technical decision-making, AI-assisted development, and production delivery. Able to explain technical workflows and trade-offs clearly to developers and non-technical stakeholders.

### TECHNICAL EXPERTISE

**Applied AI:** LLM applications, AI agents, agentic workflows, RAG, prompt design, guardrails, evaluation, AI-assisted software development

**Software Engineering:** PHP, Laravel, JavaScript, Node.js, Vue.js, REST APIs, authentication/RBAC, system integrations, document workflows

**Data & Delivery:** PostgreSQL, MySQL, Redis, Docker, Git/GitHub, CI/CD, cloud/VPS deployment, production troubleshooting

### PROFESSIONAL EXPERIENCE

**Senior Software Engineer | Universitas Sebelas Maret** Sep 2025 – Present

- Architected institutional systems for task assignments, expense workflows, and legal documents, including multilevel approvals, role-based access, and automated PDF generation.
- Implemented geo-verified reporting with photo watermarking, payment gateway integrations, data validation, and audit-oriented backend workflows.
- Own full-stack delivery from stakeholder requirements and architecture through implementation, testing, and deployment.

**Lead Software Engineer & Solutions Architect | Root Labs Indonesia** Aug 2020 – Sep 2025

- Led solution architecture and end-to-end technical delivery of client systems, from requirements and backend design to implementation, deployment, and maintenance.
- Made technical design decisions, coordinated engineering work, and balanced delivery requirements with maintainability and production stability.

**Software Engineer / Product Manager | Learn Quran** Nov 2022 – Aug 2025

- Built and maintained production APIs and full-stack applications for multi-platform Quran products, improving backend reliability, query performance, and feature delivery.
- Combined hands-on engineering with product roadmap planning, prioritization, app-store operations, and cross-functional coordination.

**Web Developer | Universitas Sebelas Maret** May 2021 – Jun 2024

- Developed faculty-wide Laravel/CodeIgniter applications and REST APIs; coordinated technical requirements with academic stakeholders.

## SELECTED ENGINEERING PROJECTS

### AsistenDiet — AI Nutrition Companion

asistendiet.com

- AI-powered nutrition companion that turns WhatsApp messages and food photos into estimated nutrition logs, daily summaries, personalized targets, and a connected web dashboard.
- Project role: Founder & Applied AI Engineer. Led product strategy, LLM workflows, backend architecture, integrations, deployment, and operations.

### BundaGizi — AI Child Nutrition Assistant

bundagizi.com

- WhatsApp-first AI assistant for complementary feeding (MPASI) and child growth, with age-aware guidance, growth tracking, immunization reminders, and web reporting.
- Project role: Founder & Applied AI Engineer. Led conversational AI integration, backend services, data workflows, deployment, and ongoing iteration.

### BidikTender — AI Tender Document Platform

bidiktender.web.id

- AI-assisted tender workspace that analyzes procurement requirements, matches them against supporting documents, flags missing evidence, and supports RAB–BOQ comparison.
- Project role: Lead AI Engineer & Solution Architect. Designed the architecture, document-analysis pipeline, validation workflows, human-review process, and production application.

### Open-Source AI Agent Tooling — JavaScript Libraries

github.com/hijrahassalam

- Reusable JavaScript tooling for AI agent applications, spanning LLM routing, caching, failover, agent memory, RAG, guardrails, evaluation, and cost management.
- Project role: Creator & Developer. Designed and published reusable components for reliable AI-agent workflows and integrations.

## RESEARCH, COMMUNITY & RECOGNITION

**Invited Speaker — Build With AI Solo 2026 (GDG Semarang):** Led a hands-on workshop on Gemini, AI-assisted development, and moving from prompts to working applications.

**Explainable Medical AI — COVID Chest X-ray Classification:** Built a TensorFlow/Keras CNN with Grad-CAM explainability; reported 95.27% accuracy on the COVID-QU-Ex benchmark. github.com/hijrahassalam/covid-cxr-gradcam

**Publication:** COVID-19 case forecasting with Single Exponential Smoothing and Holt methods (Ultimatics, 2020).

**Recognition:** Bronze Medal, Advanced Innovation JAM Japan (2019).

## EDUCATION & CERTIFICATIONS

**Universitas Sebelas Maret | Bachelor of Informatics Education, 2017–2021**

GPA: 3.86/4.00 (Cum Laude) | Undergraduate thesis: CNN-based COVID-19 detection from X-ray and CT images

EF SET English C1 Advanced | Google IT Automation with Python | Digital Talent Scholarship – Backend Developer | BNSP Programmer Certification